Experiment 7: Develop a Simple Calculator Using XML-RPC.

Aim - To develop a Simple Calculator using the XML-RPC (XML Remote Procedure Call) protocol in Python to perform basic arithmetic operations between a client and a server.

Requirements

Hardware

• Laptop/Desktop

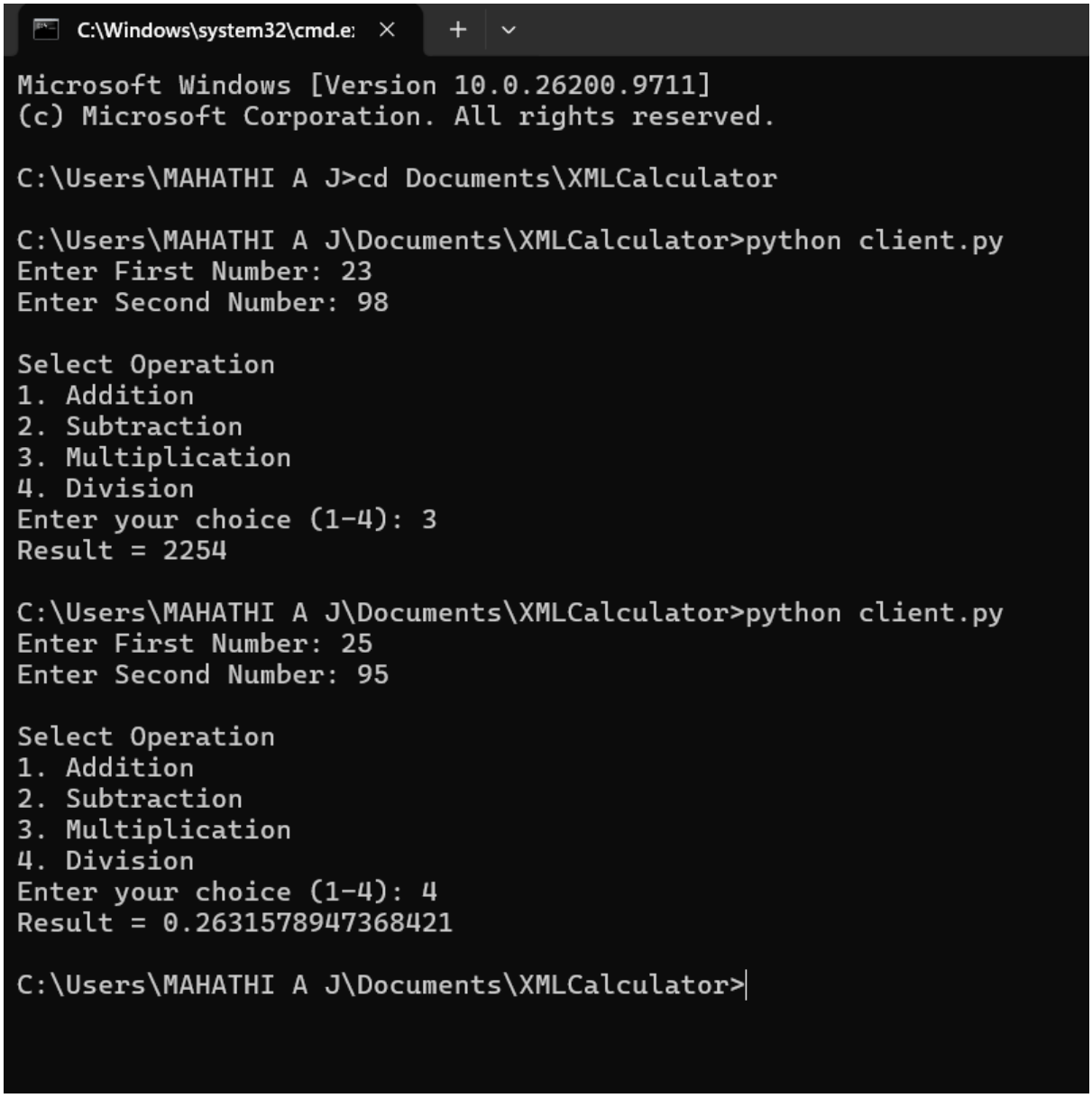
• Minimum 4 GB RAM

Software

• Windows 10/11 or Ubuntu Linux

• Python 3.8 or above

• Visual Studio Code (VS Code) / IDLE / PyCharm



server.py - C:/Users/MAHATHI A J/Documents/XMLCalculator/server.py (3.14.0)

File Edit Format Run Options Window Help

```
from xmlrpc.server import SimpleXMLRPCServer

server = SimpleXMLRPCServer(("localhost", 8000))

print("XML-RPC Calculator Server Running on Port 8000...")

def add(a, b):
    return a + b

def subtract(a, b):
    return a - b

def multiply(a, b):
    return a * b

def divide(a, b):
    if b == 0:
        return "Division by zero is not allowed."
    return a / b

server.register_function(add, "add")
server.register_function(subtract, "subtract")
server.register_function(multiply, "multiply")
server.register_function(divide, "divide")

try:
    server.serve_forever()
except KeyboardInterrupt:
    print("\nServer stopped.")
    server.server_close()
```

client.py - C:/Users/MAHATHI A J/Documents/XMLCalculator/client.py (3.14.0)

File Edit Format Run Options Window Help

```
import xmlrpc.client

server = xmlrpc.client.ServerProxy("http://localhost:8000")

a = int(input("Enter First Number: "))
b = int(input("Enter Second Number: "))

print("\nSelect Operation")
print("1. Addition")
print("2. Subtraction")
print("3. Multiplication")
print("4. Division")

choice = input("Enter your choice (1-4): ")

try:
    if choice == '1':
        print("Result =", server.add(a, b))

    elif choice == '2':
        print("Result =", server.subtract(a, b))

    elif choice == '3':
        print("Result =", server.multiply(a, b))

    elif choice == '4':
        print("Result =", server.divide(a, b))

    else:
        print("Invalid Choice")

except (xmlrpc.client.Error, OSError) as e:
    print("Could not communicate with the server:", e)
```